

ASIRI SANDAKELUM

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🌐 linkedin.com/in/asiri-sandakelum | 📁 Asiri Sandakelum Portfolio

Professional Summary

Final-year Mechanical Engineering undergraduate at the University of Moratuwa with hands-on experience in mechanical design, product development, CAD modelling, prototyping, and manufacturing systems. Experienced in SolidWorks, AutoCAD, FEA, robotics, and automation projects through industrial training and engineering design projects. Passionate about developing innovative mechanical solutions and contributing to multidisciplinary engineering teams.

Education

University of Moratuwa

BSc Eng (Hons) specialised in the field of Mechanical Engineering.

Moratuwa, Sri Lanka

2022 – Present

- **Relevant Coursework:** Manufacturing Processes, Machine Design, Mechatronics & PLC, Embedded Systems, Engineering Mathematics, Thermodynamics, Fluid Mechanics, Engineering Materials, Operations Research

Work Experience

Future Fibres Lanka (Pvt) Ltd — Maintenance Section

Mechanical Engineering Intern

Biyagama EPZ, Sri Lanka

Dec 2024 – May 2025

- Redesigned a **Bobbin Winding Tensioner** in SolidWorks, eliminating roller misalignment; improved yarn tension consistency and reduced thread breakage, verified through fabrication and live testing.
- Developed an **IoT-based Temperature & Humidity Monitoring System** (ESP32, DHT22, DS18B20) with real-time cloud logging via ThingSpeak, enabling remote environmental tracking across production zones.
- Conducted a carbon dust filtration case study for the AeroSIX plate cutting area; engineered a sealed HEPA filtration enclosure that significantly improved air quality and compliance with worker safety standards.
- Drafted AutoCAD layout drawings for floor plans, piping systems, and drainage; supported CMMS implementation and contributed to preventive maintenance scheduling and 5S standardisation programmes.
- Gained direct exposure to carbon fibre composite manufacturing processes for high-performance rigging systems distributed to global marine and sailing markets.

Key Engineering Projects [View Full Portfolio](#)

Active Knee Exoskeleton for Osteoarthritis Treatment

Ongoing

- Optimised linkage parameters using a Genetic Algorithm (GA) to replicate the human knee Instantaneous Centre of Rotation (ICR), achieving a mean tracking error of **0.47 mm**.
- Developed a decentralised Bowden cable actuation system to reduce distal limb inertia and improve wearer comfort during gait assistance.
- Implemented embedded sensing and gait-phase detection for assistive torque generation and rehabilitation-oriented control.
- **Tools:** MATLAB, Genetic Algorithm, SolidWorks

Below-Knee Prosthetic Leg - Fatigue Load Testing Mechanism

Dec 2025

- Designed and prototyped a custom mechanical motion-generation system, transforming design requirements into a functional mechanism through concept development, mechanism synthesis, and iterative design refinement.
- Applied engineering design methodologies, CAD modeling, and DFMA principles to optimize manufacturability and facilitate rapid prototype development and testing.
- **Tools:** SolidWorks, Mechanical Design, Product Development, DFMA, 3D Printing (PLA)

Operational Throughput Enhancement - Abans Auto (Pvt) Ltd

May 2026

- Identified bottlenecks in PDI/QC (75 min/unit) and batch material handling (90 min/batch) at a Hero Motorcycle Assembly Plant; proposed drive-through workstation layout, dedicated functional test track, and strategic safety stock management.

Reverse Engineering - Bicycle Disc Brake System

May 2024

- Performed full material characterisation (burn tests, density measurements, Rockwell hardness testing); developed optimised 3D models and validated performance via FEA simulation and generative design.
- **Tools:** Fusion 360, ANSYS FEA, Generative Design

Gearbox Design - Combat Robot Weapon Systems

Nov 2024

- Designed a high-torque multi-speed gearbox for vertical and horizontal spinner weapons; completed gear selection, shaft stress analysis, bearing selection, and power transmission calculations from first principles.
- **Tools:** Solid Edge, Machine Design Principles

Smart Medibox - IoT Embedded System

Aug 2024

- Built an IoT-connected medicine box monitoring temperature, humidity, and light levels with automated alerts and a real-time Node-RED dashboard; simulated and validated on Wokwi.

- *Tools:* C++, Node-RED, MQTT, Wokwi

Autonomous Car Wash Robot - Mechatronics Group Project

Jul 2023

- Led mechanical design and systems integration for a fully autonomous car wash robot; interfaced Arduino Mega and ESP8266 with a custom MIT App Inventor mobile application.
- *Tools:* SolidWorks, Arduino, MIT App Inventor

Production Line Optimisation - Aruna Tea Factory

Nov 2024

- Identified throughput bottlenecks in withering and packing sections; proposed equipment modifications and rescheduling strategies validated through simulation-based analysis.

Leadership & Volunteering

Department of Mechanical Engineering, University of Moratuwa

Student Representative

2025 – Present

- Represent student interests by serving as the primary communication channel between the department administration and the student body.
- Coordinated the dissemination of academic and administrative information across the cohort, ensuring effective student engagement.
- Supported the organisation of career development programs, industry networking events, and departmental activities.
- Collaborated with academic staff and student committees to address student concerns and improve the overall academic experience.

IEEE Robotics & Automation Society (RAS), University of Moratuwa

Vice Chairman

Nov 2024 – Oct 2025

- Directed executive decision-making and annual strategic planning for one of the university's most active IEEE student chapters.
- Mentored junior committee members in project management, event planning, and technical execution.

Co-Chair, BatTalk 2.0

Mar 2024 – Oct 2024

- Organised a flagship technical talk series on AI/ML career dynamics, coordinating with industry professionals and university faculty; managed end-to-end logistics and speaker outreach.

Sasnaka Sansada Foundation

District Coordinator — Nuwara Eliya

Mar 2024 – Dec 2024

- Managed district-level outreach operations, volunteer recruitment, and provincial seminar logistics.

Ganitha Sawiya Coordinator

May 2022 – May 2023

- Organised **50+ mathematics seminars** and educational workshops across the Nuwara Eliya District, directly improving O/L and A/L mathematics pass rates among rural school students.

EXMO 2023 — University of Moratuwa

Exhibit Coordinator

Apr 2023 – Jul 2023

- Curated and managed mechanical engineering exhibits for Sri Lanka's premier university engineering exhibition; oversaw space allocation and technical safety for thousands of visitors.

Technical Skills

CAD & Design: SolidWorks, Fusion 360, Solid Edge, AutoCAD

Simulation & Analysis: ANSYS (FEA), MATLAB

Programming: Python, C/C++, Arduino IDE

IoT & Embedded Systems: ESP32, DHT22, DS18B20, ThingSpeak, Node-RED, MQTT

Manufacturing: Basic manufacturing processes, DFM, CNC machining, sheet metal fabrication, welding, 3D printing

Engineering Software: MS Office Suite, Photoshop, DaVinci Resolve, After Effects

Languages: English (Professional Proficiency), Sinhala (Native)

Core Competencies: Mechanical design, preventive maintenance, process optimisation, technical documentation, cross-functional team leadership

References

Dr. Pubudu Ranaweera

Senior Lecturer

Department of Mechanical Engineering

University of Moratuwa, Sri Lanka

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